A compact Titans neural-memory model for DNA

Decoder-hidden diversity, stable online memory—and incomplete genomic organization

c16 exploratory study • 5M training bases • 2.52M parameters • July 2026

The question

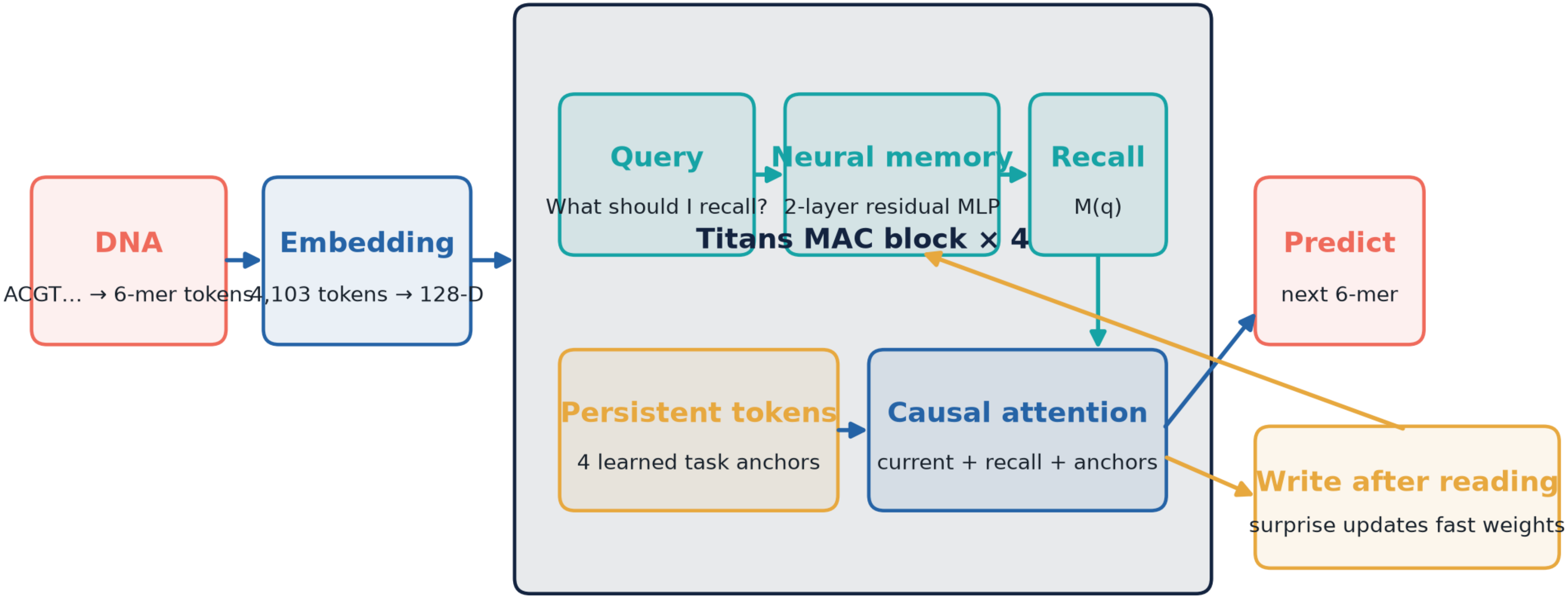
Can this compact paper-deep Titans model:

- train with exact online neural-memory updates?
- retain broad DNA sequence diversity?
- generate E. coli-like local and gene-scale organization?
- justify a larger adaptive-only discovery run?

Boundary: no matched 5M no-memory control → no memory-benefit claim.

How the tested model reads, remembers and predicts DNA

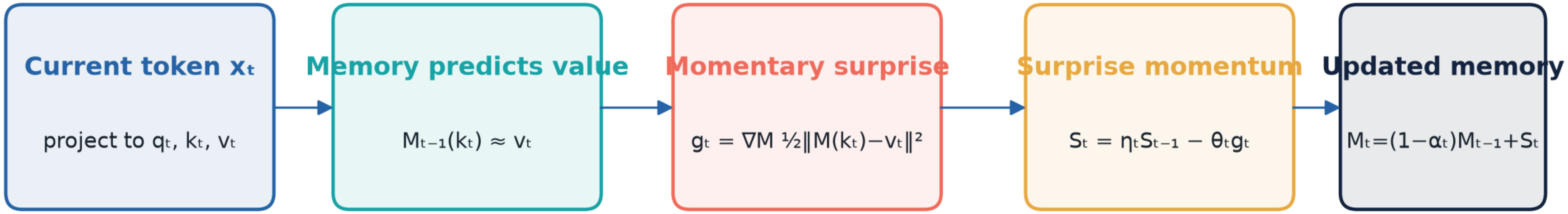
A compact Memory-as-Context (MAC) network; four identical blocks are stacked.



Memory is stream-specific and changes during generation; ordinary model weights stay fixed.

The mathematics: remember what was surprising

Neural memory update: prediction error becomes “surprise”



η: how long past surprise persists **θ: strength of the new write** **α: fraction of old memory forgotten**

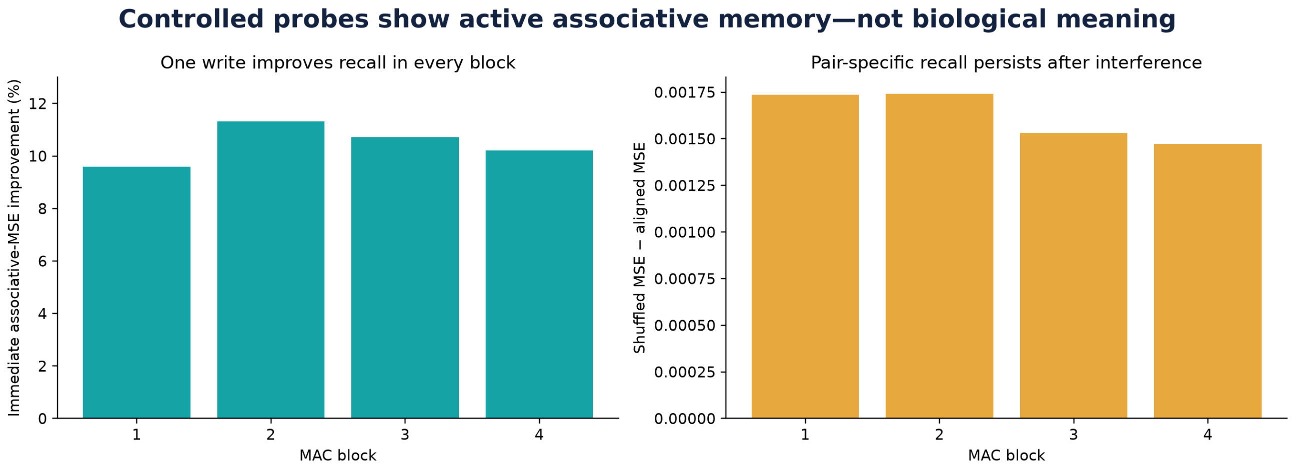
c16 initializes $\eta=0.9$, $\theta=0.001$ and $\alpha=0.001$, learns them by layer/channel, and applies no surprise cap.

Paper recurrence preserved; c16 uses exact evolving nonlinear gradients, per-channel gates, and no surprise clipping.

The model tested

2.52M	parameters
4 ×	MAC blocks
128	hidden width
2-layer	residual MLP memory
5.0M	training bases
0.0388%	of one corpus pass
1.9828	held-out BPB
T4	training GPU

Neural memory is active and stable



0%

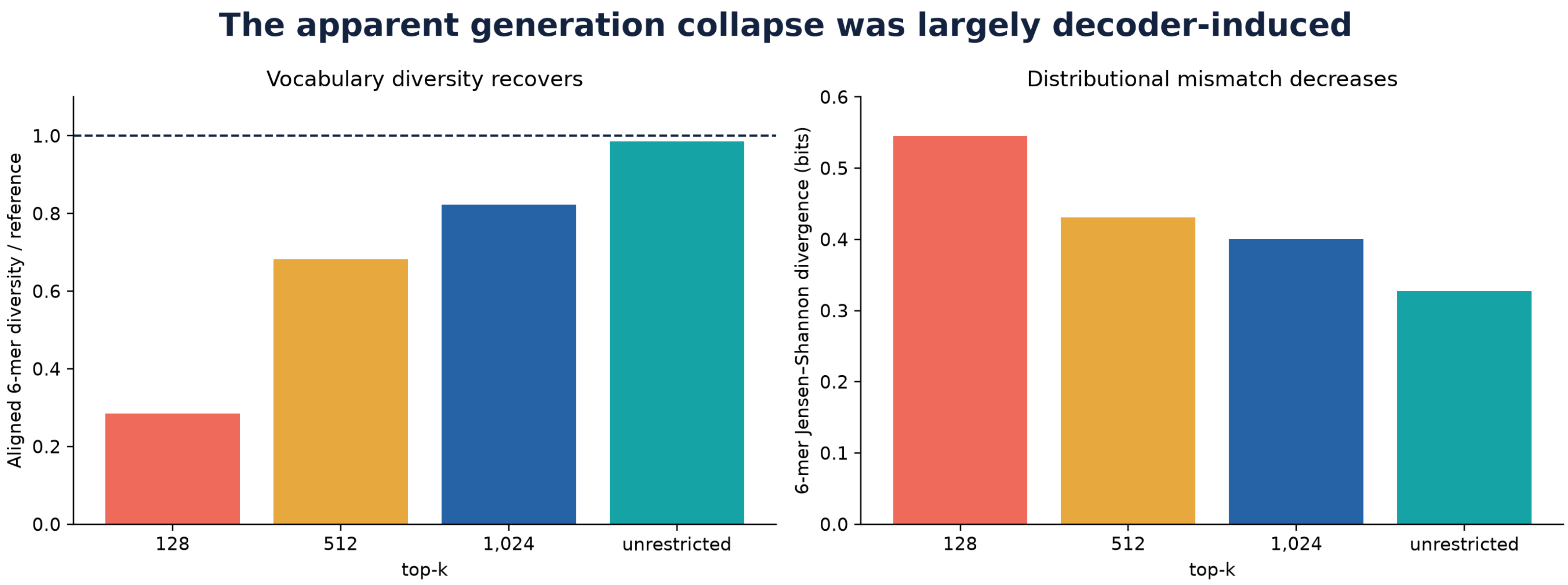
gradient interventions

4 / 4

blocks retain delayed association

Mechanism works ≠ memory improves genomics.

Key result: top-k created the apparent collapse



Temperature mattered less than vocabulary truncation

At k=1,024 and p=.99

T	diversity/ref	6-mer JSD
.6	.802	.4055
.8	.822	.4010
1.0	.838	.3939
1.1	.842	.3933

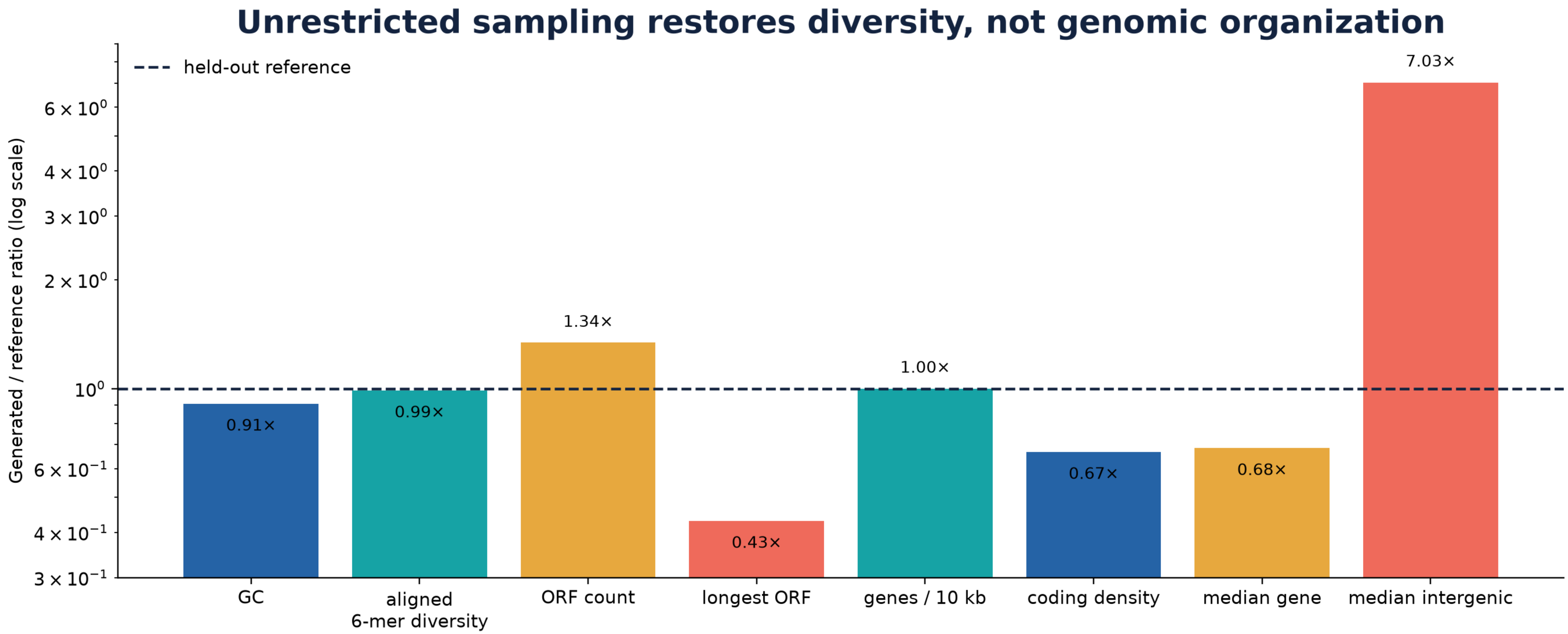
Temperature 0.6 was not advantageous.

Without top-k

p=.95 → 6-mer JSD .3335
p=.99 → .3270
p=1.0 → .3261

Freeze scientific diagnostic:
T=.8 · no top-k · p=1.0

Diversity recovered; genome organization did not



What the best samples look like statistically

Metric	reference	generated	assessment
GC	51.42%	46.71%	AT-biased
aligned diversity	.9219	.9111	near reference
ORFs ≥90 bp	28	37.5	too many
longest ORF	864 bp	372 bp	too short
Prodigal genes / 10 kb	8.14	8.14	count matches; n=5
coding density	.494	.330	33% lower
median gene	504 bp	345 bp	32% shorter
median intergenic	72 bp	506 bp	7× longer

What we may—and may not—claim

Evidence ladder: where the present study stops

Established	finite training · exact resume · active memory · broad decoded diversity
Supported hint	short-range bacterial statistics · coding-like sequence
Not established	adaptive-memory benefit · taxonomy · anomaly utility
Prohibited inference	functional genes · viability · promoter activity · safety

Next decision path

1. **Confirm decoding**
   4 prompts × 1,024 tokens; reference-v-reference baseline
2. **Freeze diverse panel**
   clade/accession-stratified validation
3. **A100 qualification**
   24.8M parameters; 12 blocks × d256
4. **Adaptive-only 25M**
   evaluate at 5M, 10M and 25M
5. **Controls only if useful**
   no-memory, frozen, shallow, second seed

Scale is justified as a controlled discovery—not as a whole-corpus or functional-generation claim.

Take-home message

The compact model learned broad short-range bacterial sequence support and stable online memory dynamics.

It has not yet learned convincing gene-scale organization—and adaptive memory has not yet been shown to cause a benefit.

Qualified green light: confirm, stratify, then scale to the 25M Medium discovery gate.